Sound: pitch and amplitude

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Learning target: Use a longitudinal-wave model to distinguish pitch and sound level.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 5; 3 course days later.*
Away from the normal; frequency stays fixed at a stationary boundary.

02. Explain a worked example (worked)
Same horizontal spacing of cycles, different vertical excursions.

03. Look and explain (model)
Compressions dense, rarefactions sparse; particles oscillate locally in basic model.

04. Build the solution (practice)
B; 400 and 800 cycles.

05. Use the idea (reason)
Amplitude/intensity changes; frequency content need not double.

06. Choose a method (mixed) *Review: Unit 5, Day 8; 12 course days later.*
Work input enables the transfer; it does not contradict the direction of spontaneous heat flow.

07. Exit check (exit)
Longitudinal; oscillate parallel.

08. STRETCH: Apply it another way (stretch)
Pitch change alters frequency/period; amplitude change changes vertical size at constant period.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 5; 3 course days later.*
$\theta_c \approx 38.68°$ from the normal. Above it, total internal reflection occurs.

02. Explain a worked example (worked)
Same horizontal spacing of cycles, different vertical excursions.

03. Represent and interpret (model)
Compressions dense, rarefactions sparse; particles oscillate locally in basic model.

04. Solve and justify (practice)
1.36 m;.68 m; intensity quadruples.

05. Analyze the reasoning (reason)
Amplitude/intensity changes; frequency content need not double.

06. Choose a method (mixed) *Review: Unit 5, Day 8; 12 course days later.*
Total energy is conserved. At common temperature the initial temperature-difference opportunity for work is eliminated.

07. Exit check (exit)
50 cycles; T=.002 s.

08. STRETCH: Apply it another way (stretch)
5 beats/s is amplitude modulation of the combined sound; original tones remain near 400 Hz.